

## CODE PART-1

```
1#include <stdio.h>
2#include <stdlib.h>
3#include <time.h>
4
5void initialize_game();
6void print_board();
7void move_player(char direction);
8
9int class_1_1_x, class_1_1_y, class_1_2_x, class_1_2_y, class_1_3_x, class_1_3_y, class_1_4_x, class_1_4_y; /* toplanmasi gereken sayilar icin x ve y koordinatlarinin tanimlanmasi */
10int class_2_1_x, class_2_1_y, class_2_2_x, class_2_2_y, class_2_3_x, class_2_3_y;
11int positionP_x = 9, positionP_y = 9; /* P nin ilk pozisyon */
12int sizeOfBoard_x, sizeOfBoard_y;
13int totalECTS = 0;
14int canPassFirstWall = 0, canPassSecondWall = 0, isPassedFirstWall = 0, isPassedSecondWall = 0; /* duvar gecisleri icin flaglar */
15char direction;
16int isGameEnd = 0;
17
18int main() {
19
20    initialize_game();
21    print_board();
22    move_player(direction);
23
24    return 0;
25}
```

In this part ; I defined functions and variables. On lines 9-10 I created separate variables for the coordinates of the numbers that need to be collected. On line 11 I defined initial positions of the P. And on line 14 I defined flags for the pass wall situation.

## CODE-PART-2

```
26void initialize_game(){
27
28    int areTheySame1 = 1, areTheySame2 = 1;
29    int minFor1 = 8, maxFor1 = 10, minFor2 = 6, maxFor2 = 12; /* duvarlarin konuma gore toplanmasi gereken konumlarin max-min koordinlari */
30    int range1 = maxFor1 - minFor1 + 1; /* random uretin icin range */
31    int range2 = maxFor2 - minFor2 + 1;
32
33    srand(time(NULL));
34
35    // min ile max arasinda rastgele bir sayi uret onlari da 1 lerin konumu yap.
36    while(areTheySame1 == 1){
37        class_1_1_x = rand() % range1 + minFor1;
38        class_1_1_y = rand() % range1 + minFor1 - 1;
39        class_1_2_x = rand() % range1 + minFor1;
40        class_1_2_y = rand() % range1 + minFor1 - 1;
41        class_1_3_x = rand() % range1 + minFor1;
42        class_1_3_y = rand() % range1 + minFor1 - 1;
43        class_1_4_x = rand() % range1 + minFor1;
44        class_1_4_y = rand() % range1 + minFor1 - 1;
45        /* uretilen noktalar birbirine esit mi diye kontrol et.*/
46        if((class_1_1_x == class_1_2_x && class_1_1_y == class_1_2_y) ||
47           (class_1_1_x == class_1_3_x && class_1_1_y == class_1_3_y) ||
48           (class_1_1_x == class_1_4_x && class_1_1_y == class_1_4_y) ||
49           (class_1_2_x == class_1_3_x && class_1_2_y == class_1_3_y) ||
50           (class_1_2_x == class_1_4_x && class_1_2_y == class_1_4_y) ||
51           (class_1_3_x == class_1_4_x && class_1_3_y == class_1_4_y) ){
52            areTheySame1 = 1;
53        }else if((class_1_1_x == 9 && class_1_1_y == 9) || /* uretilen noktalar P'nin ilk pozisyonuna esit mi (9,9) kontrol et.*/
54                (class_1_2_x == 9 && class_1_2_y == 9) ||
55                (class_1_3_x == 9 && class_1_3_y == 9) ||
56                (class_1_4_x == 9 && class_1_4_y == 9) ){
57            areTheySame1 = 1;
58        }else{
59            areTheySame1 = 0;
60        }
61    }
62}
```

In this part; in while loop I prevented the same numbers from being produced. Also I ensured that the numbers were generated in a different location than the initial position of the P.

## CODE-PART-3

```
63 while(areTheySame2 == 1){
64     class_2_1_x = rand() % range2 + minFor2;
65     class_2_2_x = rand() % range2 + minFor2;
66     class_2_3_x = rand() % range2 + minFor2;
67
68     // 1. sayı için yap
69     if(class_2_1_x == 6 || class_2_1_x == 12 ){ /* sayının satir degeri 6 veya 12 ise y degeri 6 ile 12 arasında bir dger alabilir.*/
70         class_2_1_y = rand() % range2 + minFor2 - 1;
71     }else{ /* sayının satir degeri 6 veya 12 degil ise y degeri 5 ya da 11 degerlerini alabilir.*/
72         int sixOrTwelve = rand() % 2; /* 0 veya 1 sayisi uret. 0 ise y degeri 6 olacak 1 ise y degeri 12 olacak.*/
73         if(sixOrTwelve == 0){
74             class_2_1_y = 5;
75         }else{
76             class_2_1_y = 11;
77         }
78     }
79
80     // ikinci sayı için yap
81     if(class_2_2_x == 6 || class_2_2_x == 12 ){ /* sayının satir degeri 6 veya 12 ise y degeri 6 ile 12 arasında bir dger alabilir.*/
82         class_2_2_y = rand() % range2 + minFor2 - 1;
83     }else{
84         int sixOrTwelve = rand() % 2; /* 0 veya 1 sayisi uret. 0 ise y degeri 5 olacak 1 ise y degeri 11 olacak.*/
85         if(sixOrTwelve == 0){
86             class_2_2_y = 5;
87         }else{
88             class_2_2_y = 11;
89         }
90     }
91
92     // ucuncu sayı için yap
93     if(class_2_3_x == 6 || class_2_3_x == 12 ){ /* sayının satir degeri 6 veya 12 ise y degeri 6 ile 12 arasında bir dger alabilir.*/
94         class_2_3_y = rand() % range2 + minFor2 - 1;
95     }else{ /* sayının satir degeri 6 veya 12 degil ise y degexri 5 ya da 11 degerlerini alabilir.*/
96         int sixOrTwelve = rand() % 2; /* 0 veya 1 sayisi uret. 0 ise y degeri 5 olacak 1 ise y degeri 11 olacak.*/
97         if(sixOrTwelve == 0){
98             class_2_3_y = 5;
99         }else{
100             class_2_3_y = 11;
101         }
102     }
103 }
```

In this part ; I generated coordinates of the class\_2s. If the row value is not 6 or 12, points can be placed in the 5<sup>th</sup> or 11<sup>th</sup> column.

#### CODE-PART-4

```
100         class_2_3_y = 11;
101     }
102
103 }
104
105 /* uretilen noktalar birbirine esit mi diye kontrol et.*/
106 if((class_2_1_x == class_2_2_x && class_2_1_y == class_2_2_y) ||
107    (class_2_1_x == class_2_3_x && class_2_1_y == class_2_3_y) ||
108    (class_2_2_x == class_2_3_x && class_2_2_y == class_2_3_y) ){
109     areTheySame2 = 1;
110 }else{
111     areTheySame2 = 0;
112 }
113 }
114
115 /* size of board belirle. */
116 sizeOfBoard_x = 16;
117 sizeOfBoard_y = 16;
118 }
```

In this part; I prevented the same numbers from being produced.

#### CODE-PART-5

```
120 void print_board(){
121     int lineOfBoard = 1; /* o an yazdirilan satir degiskeni */
122     char line1[16] = {'.','.', '.', '.', '.', '.', '.', '.', '.', '.', '.', '.', '.', '.', '.', '.'}; /* line 1-2-3-4-14-15-16*/
123     char line5[16] = {'.','.', '.', '.', '.', '.', '.', '.', '.', '.', '.', '.', '.', '.', '.', '.'}; /*line5-13 */
124     char line6[16] = {'.','.', '.', '.', '.', '.', '.', '.', '.', '.', '.', '.', '.', '.', '.', '.'}; /* line6-12 */
125     char line7[16] = {'.','.', '.', '.', '.', '.', '.', '.', '.', '.', '.', '.', '.', '.', '.', '.'}; /* line7-11*/
126     char line8[16] = {'.','.', '.', '.', '.', '.', '.', '.', '.', '.', '.', '.', '.', '.', '.', '.'}; /*line 8-9-10 */
127
128     /* print line1 to line4 */
129     for(int line = 1; line <= 4 ; line++){
130         for(int i = 0; i<16 ; i++){
131             if(line == positionP_x && i == positionP_y - 1){ /* ilk 4 satir icin noktalarin pozisyon P'ye esitse P yazdir.*/
132                 printf("P");
133             }else{
134                 printf("%c" ,line1[i]);
135             }
136
137             printf(" ");
138         }
139         lineOfBoard++; /* her cizimden sonra lineOfBoard 1 arttirildi*/
140         printf("\n");
141     }
142
143     /* print line5 */
144     for(int j = 0; j<16 ; j++){
145         if((lineOfBoard == positionP_x && j == positionP_y - 1) && (line5[j] != '#')){ /* noktanin pozisyon P'nin pozisyona esitse ve duvar degilse*/
146             printf("P");
147         }else if((lineOfBoard == positionP_x && j == positionP_y - 1) && (line5[j] == '#')){ /* noktanin pozisyon P'nin pozisyona esitse ve duvarsa*/
148
149         }else{
150             printf("%c" ,line5[j]);
151         }
152
153         printf(" ");
154     }
155     lineOfBoard++;
156     printf("\n");
```

In this part; I print the rows to the screen according to the arrays I define on lines from 122-126. If the generated point is equal to the position of the P. Then I print P. I incremented the lineOfBoard variable by the 1 after each line printing.

## CODE-PART-6

```
157 |
158 | /* print line6 */
159 | for(int k = 0 ; k < 16 ; k++){
160 |     if(lineOfBoard == positionP_x && k == positionP_y - 1){
161 |         printf("P");
162 |     }
163 |     else if(class_2_1_x == lineOfBoard && class_2_1_y == k ||
164 |         class_2_2_x == lineOfBoard && class_2_2_y == k ||
165 |         class_2_3_x == lineOfBoard && class_2_3_y == k
166 |     ){
167 |         printf("2");
168 |     }else{
169 |         printf("%c" ,line6[k]);
170 |     }
171 |     printf(" ");
172 | }
173 | lineOfBoard++;
174 | printf("\n");
175 |
176 | /* print line7 */
177 | for(int l = 0 ; l < 16 ; l++){
178 |     if(lineOfBoard == positionP_x && l == positionP_y - 1){
179 |         printf("P");
180 |     }
181 |     else if(class_2_1_x == lineOfBoard && class_2_1_y == l || /*class2leri ekrana yaz */
182 |         class_2_2_x == lineOfBoard && class_2_2_y == l ||
183 |         class_2_3_x == lineOfBoard && class_2_3_y == l
184 |     ){
185 |         printf("2");
186 |     }else{
187 |         printf("%c" ,line7[l]);
188 |     }
189 |     printf(" ");
190 | }
191 | lineOfBoard++;
192 | printf("\n");
193 |
```

In this part; I continued printing to the screen.

## CODE-PART-7

```
193
194  /* print line8 to line10 */
195  for(int line = 8; line <= 10 ; line++){
196      for(int i = 0; i<16 ; i++){
197
198          if(line == positionP_x && i == positionP_y - 1){ /* Initial P location.*/
199              printf("P");
200          }else if(class_1_1_x == lineOfBoard && i == class_1_1_y || /* 1 leri ekrana yazdir. */
201                  class_1_2_x == lineOfBoard && i == class_1_2_y ||
202                  class_1_3_x == lineOfBoard && i == class_1_3_y ||
203                  class_1_4_x == lineOfBoard && i == class_1_4_y ){
204
205              printf("1");
206          }else{
207              if(i == 5 || i == 11){
208                  if( class_2_1_x == lineOfBoard && class_2_1_y == i ||
209                     class_2_2_x == lineOfBoard && class_2_2_y == i ||
210                     class_2_3_x == lineOfBoard && class_2_3_y == i){
211                      printf("2");
212                  }else{
213                      printf("%c" ,line8[i]);
214                  }
215              }else{
216                  printf("%c" ,line8[i]);
217              }
218          }
219          printf(" ");
220      }
221      lineOfBoard++;
222      printf("\n");
223  }
```

In this part; I continued printing to the screen.

## CODE-PART-8

```
223     }
224
225     /* print line11 */
226     for(int i = 0 ; i < 16 ; i++){
227         if(lineOfBoard == positionP_x && i == positionP_y - 1){
228             printf("P"); /* p denk gelirse p'yi yazdir */
229         }
230         else if(class_2_1_x == lineOfBoard && class_2_1_y == i ||
231             class_2_2_x == lineOfBoard && class_2_2_y == i ||
232             class_2_3_x == lineOfBoard && class_2_3_y == i
233         ){
234             printf("2");
235         }else{
236             printf("%c" ,line7[i]);
237         }
238
239         printf(" ");
240     }
241     lineOfBoard++;
242     printf("\n");
243
244     /* print line12 */
245     for(int i = 0 ; i < 16 ; i++){
246         if(lineOfBoard == positionP_x && i == positionP_y - 1){
247             printf("P");
248         }
249         else if(class_2_1_x == lineOfBoard && class_2_1_y == i ||
250             class_2_2_x == lineOfBoard && class_2_2_y == i ||
251             class_2_3_x == lineOfBoard && class_2_3_y == i
252         ){
253             printf("2");
254         }else{
255             printf("%c" ,line6[i]);
256         }
257
258         printf(" ");
259     }
260     lineOfBoard++;
261     printf("\n");
262 }
```

In this part; I continued printing to the screen.

## CODE-PART-9

```
262
263  /* print line13 */
264  for(int i = 0 ; i < 16 ; i++){
265      if(lineOfBoard == positionP_x && i == positionP_y - 1){
266          printf("P");
267      }else{
268          printf("%c" ,line5[i]);
269      }
270      printf(" ");
271  }
272
273  lineOfBoard++;
274  printf("\n");
275
276  /* print line14 to line16 */
277  for(int line = 14; line <= 16 ; line++){
278      for(int i = 0; i<16 ; i++){
279          if(line == positionP_x && i == positionP_y - 1){ /* noktalarin pozisyon P'ye esitse P yazdir.*/
280              printf("P");
281          }else{
282              if(line==16 && i == 15 ){
283                  printf("X");
284                  lineOfBoard--;
285              }else{
286                  printf("%c" ,line1[i]);
287              }
288          }
289          printf(" ");
290      }
291      lineOfBoard++;
292      printf("\n");
293  }
294
295  /* ilk calistirma icin.*/
296  printf("Total ECTS: %d \n" ,totalECTS);
297  printf("Enter your move: ");
298  scanf("%c" , &direction);
299
300
301
```

In this part; I continued printing to the screen.

## CODE-PART-10

```
302
303 void move_player(char direction){
304     if(direction == 'A' || direction == 'a'){
305         /* ilk duvar icin */
306         if(positionP_y > 1){
307             if(positionP_y == 8){ /* icteki duvarin dibindeyse ve duvari gecebilirse*/
308                 if(canPassFirstWall == 1 ){
309                     if(isPassedFirstWall == 0){
310                         positionP_y -= 2;
311                         isPassedFirstWall = 1;
312                     }else{
313                         positionP_y -= 1;
314                     }
315                 }else{
316                     printf("\nYou don't have enough points to pass the wall! \n");
317                 }
318             }else{
319                 /* ikinci duvar icin*/
320                 if(positionP_y == 6){ /* distaki duvarin dibindeyse ve duvari gecebilirse*/
321                     if(canPassSecondWall == 1 ){
322                         if(isPassedSecondWall == 0){
323                             positionP_y -= 2;
324                             isPassedSecondWall = 1;
325                         }else{
326                             positionP_y -= 1;
327                         }
328                     }else{
329                         printf("\nYou don't have enough points to pass the wall! \n");
330                     }
331                 }else{
332                     positionP_y -= 1;
333                 }
334             }
335         }
336     }
```

In this part; I started moving the character. This is where I control wall passes.



## CODE-PART-11

```
337      /* class_1 ler icin puan toplama */
338      if(positionP_x == class_1_1_x && positionP_y - 1 == class_1_1_y ){
339          totalECTS += 8;
340          if(totalECTS == 32){
341              canPassFirstWall = 1;
342          }
343          class_1_1_x = 99; // farkli bir sayi yaptim
344          class_1_1_y = 99;
345      }else if(positionP_x == class_1_2_x && positionP_y - 1 == class_1_2_y){
346          totalECTS += 8;
347          if(totalECTS == 32){
348              canPassFirstWall = 1;
349          }
350          class_1_2_x = 99; // farkli bir sayi yaptim
351          class_1_2_y = 99;
352      }else if(positionP_x == class_1_3_x && positionP_y - 1 == class_1_3_y){
353          totalECTS += 8;
354          if(totalECTS == 32){
355              canPassFirstWall = 1;
356          }
357          class_1_3_x = 99; // farkli bir sayi yaptim
358          class_1_3_y = 99;
359      }else if(positionP_x == class_1_4_x && positionP_y - 1 == class_1_4_y){
360          totalECTS += 8;
361          if(totalECTS == 32){
362              canPassFirstWall = 1;
363          }
364          class_1_4_x = 99; // farkli bir sayi yaptim
365          class_1_4_y = 99;
366      }
367  }
```

In this part; I checked the number collection operations. If the number (class1 or class-2) was collected then I changed the coordinates of the numbers. And I increased the totalECTS. If totalECTS equal to 32 then character can pass the first wall.

## CODE-PART-12

```
367
368     /* class_2 ler icin puan toplama */
369     if(positionP_x == class_2_1_x && positionP_y - 1 == class_2_1_y ){
370         totalECTS += 8;
371         if(totalECTS == 56){
372             canPassSecondWall = 1;
373         }
374         class_2_1_x = 99; // farkli bir sayi yaptim
375         class_2_1_y = 99;
376     }else if(positionP_x == class_2_2_x && positionP_y - 1 == class_2_2_y){
377         totalECTS += 8;
378         if(totalECTS == 56){
379             canPassSecondWall = 1;
380         }
381         class_2_2_x = 99; // farkli bir sayi yaptim
382         class_2_2_y = 99;
383     }else if(positionP_x == class_2_3_x && positionP_y - 1 == class_2_3_y){
384         totalECTS += 8;
385         if(totalECTS == 56){
386             canPassSecondWall = 1;
387         }
388         class_2_3_x = 99; // farkli bir sayi yaptim
389         class_2_3_y = 99;
390     }
391
392     print_board();
393
```

## CODE-PART-13

```
393
394 }else if(direction == 'D' || direction == 'd'){
395     if(positionP_y < 16){
396
397         /* ilk duvar icin */
398         if(positionP_y == 10){
399             if(canPassFirstWall == 1){ /* icteki duvarin dibindeyse ve duvari gecebilsir*/
400
401                 if(isPassedFirstWall == 0){
402                     positionP_y += 2;
403                     isPassedFirstWall = 1;
404                 }else{
405                     positionP_y += 1;
406                 }
407
408             }else{
409                 printf("\nYou don't have enough points to pass the wall! \n");
410             }
411         }else{
412             /* ikinci duvar icin*/
413             if(positionP_y == 12){ /* distaki duvarin dibindeyse ve duvari gecebilsir*/
414                 if(canPassSecondWall == 1 ){
415                     if(isPassedSecondWall == 0){
416                         positionP_y += 2;
417                         isPassedSecondWall = 1;
418                     }else{
419                         positionP_y += 1;
420                     }
421
422                 }else{
423                     printf("\nYou don't have enough points to pass the wall! \n");
424                 }
425             }else{
426                 positionP_y += 1;
427             }
428         }
429     }
430 }
```

## CODE-PART-14

```
430
431
432     /* class_1 ler icin puan toplama */
433     if(positionP_x == class_1_1_x && positionP_y - 1 == class_1_1_y ){
434         totalECTS += 8;
435         if(totalECTS == 32){
436             canPassFirstWall = 1;
437         }
438         class_1_1_x = 99; // farkli bir sayi yaptim
439         class_1_1_y = 99;
440     }else if(positionP_x == class_1_2_x && positionP_y - 1 == class_1_2_y){
441         totalECTS += 8;
442         if(totalECTS == 32){
443             canPassFirstWall = 1;
444         }
445         class_1_2_x = 99; // farkli bir sayi yaptim
446         class_1_2_y = 99;
447     }else if(positionP_x == class_1_3_x && positionP_y - 1 == class_1_3_y){
448         totalECTS += 8;
449         if(totalECTS == 32){
450             canPassFirstWall = 1;
451         }
452         class_1_3_x = 99; // farkli bir sayi yaptim
453         class_1_3_y = 99;
454     }else if(positionP_x == class_1_4_x && positionP_y - 1 == class_1_4_y){
455         totalECTS += 8;
456         if(totalECTS == 32){
457             canPassFirstWall = 1;
458         }
459         class_1_4_x = 99; // farkli bir sayi yaptim
460         class_1_4_y = 99;
461     }
462
```

## CODE-PART-15

```
464
465     /* class_2 ler icin puan toplama */
466     if(positionP_x == class_2_1_x && positionP_y - 1 == class_2_1_y ){
467         totalECTS += 8;
468         if(totalECTS == 56){
469             canPassSecondWall = 1;
470         }
471         class_2_1_x = 99; // farkli bir sayi yaptim
472         class_2_1_y = 99;
473     }else if(positionP_x == class_2_2_x && positionP_y - 1 == class_2_2_y){
474         totalECTS += 8;
475         if(totalECTS == 56){
476             canPassSecondWall = 1;
477         }
478         class_2_2_x = 99; // farkli bir sayi yaptim
479         class_2_2_y = 99;
480     }else if(positionP_x == class_2_3_x && positionP_y - 1 == class_2_3_y){
481         totalECTS += 8;
482         if(totalECTS == 56){
483             canPassSecondWall = 1;
484         }
485         class_2_3_x = 99; // farkli bir sayi yaptim
486         class_2_3_y = 99;
487     }
488
489     print_board();
490     if(positionP_y == 16 && positionP_x == 16){
491         printf("\nYOU WON !\n");
492         isGameEnd = 1;
493     }
```

In parts of 11-12-13-14 I did same thins. In this part, I also checked whether the game was over or not.

## CODE-PART-16

```
494 }else if(direction == 'S' || direction == 's'){
495
496     if(positionP_x < 16){
497         /* ilk duvar icin*/
498         if(positionP_x == 10 ){
499             if(canPassFirstWall == 1 ){ /* icteki duvarin dibindeyse ve duvari gecebilsir*/
500                 if(isPassedFirstWall == 0){
501                     positionP_x += 2;
502                     isPassedFirstWall = 1;
503                 }else{
504                     positionP_x += 1;
505                 }
506             }else{
507                 printf("\nYou don't have enough points to pass the wall! \n");
508             }
509         }else{
510             /* ikinci duvar icin*/
511             if(positionP_x == 12){ /* distaki duvarin dibindeyse ve duvari gecebilsir*/
512                 if(canPassSecondWall == 1 ){
513                     if(isPassedSecondWall == 0){
514                         positionP_x += 2;
515                         isPassedSecondWall = 1;
516                     }else{
517                         positionP_x += 1;
518                     }
519                 }else{
520                     printf("\nYou don't have enough points to pass the wall! \n");
521                 }
522             }else{
523                 positionP_x += 1;
524             }
525         }
526     }
527 }
```

## CODE-PART-17

```
527     }
528     /* class_1 ler icin puan toplama */
529     if(positionP_x == class_1_1_x && positionP_y - 1 == class_1_1_y ){
530         totalECTS += 8;
531         if(totalECTS == 32){
532             canPassFirstWall = 1;
533         }
534         class_1_1_x = 99; // farkli bir sayi yaptim
535         class_1_1_y = 99;
536     }else if(positionP_x == class_1_2_x && positionP_y - 1 == class_1_2_y){
537         totalECTS += 8;
538         if(totalECTS == 32){
539             canPassFirstWall = 1;
540         }
541         class_1_2_x = 99; // farkli bir sayi yaptim
542         class_1_2_y = 99;
543     }else if(positionP_x == class_1_3_x && positionP_y - 1 == class_1_3_y){
544         totalECTS += 8;
545         if(totalECTS == 32){
546             canPassFirstWall = 1;
547         }
548         class_1_3_x = 99; // farkli bir sayi yaptim
549         class_1_3_y = 99;
550     }else if(positionP_x == class_1_4_x && positionP_y - 1 == class_1_4_y){
551         totalECTS += 8;
552         if(totalECTS == 32){
553             canPassFirstWall = 1;
554         }
555         class_1_4_x = 99; // farkli bir sayi yaptim
556         class_1_4_y = 99;
557     }
558 }
```

## CODE-PART-18

```
559
560      /* class_2 ler icin puan toplama */
561      if(positionP_x == class_2_1_x && positionP_y - 1 == class_2_1_y ){
562          totalECTS += 8;
563          if(totalECTS == 56){
564              canPassSecondWall = 1;
565          }
566          class_2_1_x = 99; // farkli bir sayi yaptim
567          class_2_1_y = 99;
568      }else if(positionP_x == class_2_2_x && positionP_y - 1 == class_2_2_y){
569          totalECTS += 8;
570          if(totalECTS == 56){
571              canPassSecondWall = 1;
572          }
573          class_2_2_x = 99; // farkli bir sayi yaptim
574          class_2_2_y = 99;
575      }else if(positionP_x == class_2_3_x && positionP_y - 1 == class_2_3_y){
576          totalECTS += 8;
577          if(totalECTS == 56){
578              canPassSecondWall = 1;
579          }
580          class_2_3_x = 99; // farkli bir sayi yaptim
581          class_2_3_y = 99;
582      }
583
584      print_board();
585
586      /*oyun bitti mi kontrol */
587      if(positionP_y == 16 && positionP_x == 16){
588          printf("\nYOU WON !\n");
589          isGameEnd = 1;
590          return;
591      }
```



## CODE-PART-19

```
592 }else if(direction == 'W' || direction == 'w'){
593     if(positionP_x > 1){
594         /* ilk duvar icin */
595         if(positionP_x == 8 ){
596             if(canPassFirstWall == 1){ /* icteki duvarin dibindeyse ve duvari gecebilirse*/
597                 if(isPassedFirstWall == 0){
598                     positionP_x -= 2;
599                     isPassedFirstWall = 1;
600                 }else{
601                     positionP_x -= 1;
602                 }
603             }else{
604                 printf("\nYou don't have enough points to pass the wall! \n");
605             }
606         }else{
607             /* ikinci duvar icin*/
608             if(positionP_x == 6){ /* distaki duvarin dibindeyse ve duvari gecebilirse*/
609                 if(canPassSecondWall == 1 ){
610                     if(isPassedSecondWall == 0){
611                         positionP_x -= 2;
612                         isPassedSecondWall = 1;
613                     }else{
614                         positionP_x -= 1;
615                     }
616                 }else{
617                     printf("\nYou don't have enough points to pass the wall! \n");
618                 }
619             }else{
620                 positionP_x -= 1;
621             }
622         }
623     }
624 }
625
626 }
627 }
```

## CODE-PART-20

```
628      /* class_1 ler icin puan toplama */
629      if(positionP_x == class_1_1_x && positionP_y - 1 == class_1_1_y ){
630          totalECTS += 8;
631          if(totalECTS == 32){
632              canPassFirstWall = 1;
633          }
634          class_1_1_x = 99; // farkli bir sayi yaptim
635          class_1_1_y = 99;
636      }else if(positionP_x == class_1_2_x && positionP_y - 1 == class_1_2_y){
637          totalECTS += 8;
638          if(totalECTS == 32){
639              canPassFirstWall = 1;
640          }
641          class_1_2_x = 99; // farkli bir sayi yaptim
642          class_1_2_y = 99;
643      }else if(positionP_x == class_1_3_x && positionP_y - 1 == class_1_3_y){
644          totalECTS += 8;
645          if(totalECTS == 32){
646              canPassFirstWall = 1;
647          }
648          class_1_3_x = 99; // farkli bir sayi yaptim
649          class_1_3_y = 99;
650      }else if(positionP_x == class_1_4_x && positionP_y - 1 == class_1_4_y){
651          totalECTS += 8;
652          if(totalECTS == 32){
653              canPassFirstWall = 1;
654          }
655          class_1_4_x = 99; // farkli bir sayi yaptim
656          class_1_4_y = 99;
657      }
658      ---
```

## CODE-PART-21

```
660      /* class_2 ler icin puan toplama */
661      if(positionP_x == class_2_1_x && positionP_y - 1 == class_2_1_y ){
662          totalECTS += 8;
663          if(totalECTS == 56){
664              canPassSecondWall = 1;
665          }
666          class_2_1_x = 99; // farkli bir sayi yaptim
667          class_2_1_y = 99;
668      }else if(positionP_x == class_2_2_x && positionP_y - 1 == class_2_2_y){
669          totalECTS += 8;
670          if(totalECTS == 56){
671              canPassSecondWall = 1;
672          }
673          class_2_2_x = 99; // farkli bir sayi yaptim
674          class_2_2_y = 99;
675      }else if(positionP_x == class_2_3_x && positionP_y - 1 == class_2_3_y){
676          totalECTS += 8;
677          if(totalECTS == 56){
678              canPassSecondWall = 1;
679          }
680          class_2_3_x = 99; // farkli bir sayi yaptim
681          class_2_3_y = 99;
682      }
683      print_board();
684  }else{
685      printf("Enter a valid direction. \n");
686  }
687
688  while(isGameEnd == 0){
689      printf("Enter your move: ");
690      scanf("%c", &direction);
691      move_player(direction);
692  }
693
694 }
695
```

I did some things for 'W', 'A', 'S', 'D'. And end of the this function I continue the movement of the process as long as the game does not end.

## OUTPUT-PART-1

```

. . . . # . # . P . # . # . . .
. . . . # . # 1 1 1 # . # . . .
. . . . # . # # # # # . # . . .
. . . . # . . . . . . # . . .
. . . . # # # # # # # # . . .
. . . . . . . . . . . . . . .
. . . . . . . . . . . . . . .
. . . . . . . . . . . . . . . X
Total ECTS: 0
Enter your move: d
. . . . . . . . . . . . . . .
. . . . . . . . . . . . . . .
. . . . . . . . . . . . . . .
. . . . . . . . . . . . . . .
. . . . # # # # # # # # . . .
. . . . # . 2 . . . . # . . .
. . . . # . # # # # # 2 # . . .
. . . . # . # . . 1 # 2 # . . .
. . . . # . # . . P # . # . . .
. . . . # . # 1 1 1 # . # . . .
. . . . # . # # # # # . # . . .
. . . . # . . . . . . # . . .
. . . . # # # # # # # # . . .
. . . . . . . . . . . . . . .
. . . . . . . . . . . . . . .
. . . . . . . . . . . . . . . X
Total ECTS: 0
Enter your move: Enter your move: d

You don't have enough points to pass the wall!
. . . . . . . . . . . . . . .
. . . . . . . . . . . . . . .
. . . . . . . . . . . . . . .
. . . . . . . . . . . . . . .
. . . . # # # # # # # # . . .
. . . . # . 2 . . . . # . . .
. . . . # . # # # # # 2 # . . .
. . . . # . # . . 1 # 2 # . . .
. . . . # . # . . P # . # . . .
. . . . # . # 1 1 1 # . # . . .
. . . . # . # # # # # . # . . .
. . . . # . . . . . . # . . .
. . . . # # # # # # # # . . .
. . . . . . . . . . . . . . .
. . . . . . . . . . . . . . .
. . . . . . . . . . . . . . . X
Total ECTS: 0
Enter your move: Enter your move:

```

## OUTPUT-PART-2

[illegible]

### OUTPUT-PART-3

```

. . . . # . # # # # 2 # . . .
. . . . # . # . P 1 # 2 # . . .
. . . . # . # . . . # . # . . .
. . . . # . # . . . # . # . . .
. . . . # . # # # # # . # . . .
. . . . # . . . . . . # . . .
. . . . # # # # # # # # # . . .
. . . . . . . . . . . . . . .
. . . . . . . . . . . . . . .
. . . . . . . . . . . . . . X
Total ECTS: 24
Enter your move: Enter your move: d
. . . . . . . . . . . . . . .
. . . . . . . . . . . . . . .
. . . . . . . . . . . . . . .
. . . . . . . . . . . . . . .
. . . . # # # # # # # # # . . .
. . . . # . 2 . . . . . # . . .
. . . . # . # # # # # 2 # . . .
. . . . # . # . . P # 2 # . . .
. . . . # . # . . . # . # . . .
. . . . # . # . . . # . # . . .
. . . . # . # # # # # . # . . .
. . . . # . . . . . . # . . .
. . . . # # # # # # # # # . . .
. . . . . . . . . . . . . . .
. . . . . . . . . . . . . . .
. . . . . . . . . . . . . . X
Total ECTS: 32
Enter your move: Enter your move: d
. . . . . . . . . . . . . . .
. . . . . . . . . . . . . . .
. . . . . . . . . . . . . . .
. . . . . . . . . . . . . . .
. . . . # # # # # # # # # . . .
. . . . # . 2 . . . . . # . . .
. . . . # . # # # # # 2 # . . .
. . . . # . # . . . # P # . . .
. . . . # . # . . . # . # . . .
. . . . # . # . . . # . # . . .
. . . . # . # # # # # . # . . .
. . . . # . . . . . . # . . .
. . . . # # # # # # # # # . . .
. . . . . . . . . . . . . . .
. . . . . . . . . . . . . . .
. . . . . . . . . . . . . . X
Total ECTS: 40
Enter your move: Enter your move:
```

## OUTPUT-PART-4

```

. . . . # . # . . . # . # . . .
. . . . # . # . . . # . # . . .
. . . . # . # # # # # . # . . .
. . . . # . . . . . . # . . .
. . . . # # # # # # # # . . .
. . . . . . . . . . . . . . .
. . . . . . . . . . . . . . .
. . . . . . . . . . . . . . X
Total ECTS: 48
Enter your move: Enter your move: a
. . . . . . . . . . . . . . .
. . . . . . . . . . . . . . .
. . . . . . . . . . . . . . .
. . . . . . . . . . . . . . .
. . . . # # # # # # # # . . .
. . . . # . 2 . P . . . # . . .
. . . . # . # # # # # . # . . .
. . . . # . # . . . # . # . . .
. . . . # . # . . . # . # . . .
. . . . # . # . . . # . # . . .
. . . . # . # # # # # . # . . .
. . . . # . . . . . . # . . .
. . . . # # # # # # # # . . .
. . . . . . . . . . . . . . .
. . . . . . . . . . . . . . .
. . . . . . . . . . . . . . X
Total ECTS: 48
Enter your move: Enter your move: a
. . . . . . . . . . . . . . .
. . . . . . . . . . . . . . .
. . . . . . . . . . . . . . .
. . . . . . . . . . . . . . .
. . . . # # # # # # # # . . .
. . . . # . 2 P . . . # . . .
. . . . # . # # # # # . # . . .
. . . . # . # . . . # . # . . .
. . . . # . # . . . # . # . . .
. . . . # . # . . . # . # . . .
. . . . # . # # # # # . # . . .
. . . . # . . . . . . # . . .
. . . . # # # # # # # # . . .
. . . . . . . . . . . . . . .
. . . . . . . . . . . . . . X
Total ECTS: 48
Enter your move: Enter your move: w

You don't have enough points to pass the wall!
```

## OUTPUT-PART-5

[illegible]



## OUTPUT-PART-6

```

. . . . # . # . . . # . # . . .
. . . . # . # . . . # . # . . .
. . . . # . # # # # # . # . . .
. . . . # . . . . . # . . . .
. . . . # # # # # # # # . . .
. . . . . . . . . . . . . . P
. . . . . . . . . . . . . .
. . . . . . . . . . . . . . X
Total ECTS: 56
Enter your move: Enter your move: s
. . . . . . . . . . . . . .
. . . . . . . . . . . . . .
. . . . . . . . . . . . . .
. . . . . . . . . . . . . .
. . . . # # # # # # # # . . .
. . . . # . . . . . # . . .
. . . . # . # # # # # . # . . .
. . . . # . # . . . # . # . . .
. . . . # . # . . . # . # . . .
. . . . # . # . . . # . # . . .
. . . . # . # # # # # . # . . .
. . . . # . . . . . # . . .
. . . . # # # # # # # # . . .
. . . . . . . . . . . . . .
. . . . . . . . . . . . . . P
. . . . . . . . . . . . . . X
Total ECTS: 56
Enter your move: Enter your move: s
. . . . . . . . . . . . . .
. . . . . . . . . . . . . .
. . . . . . . . . . . . . .
. . . . . . . . . . . . . .
. . . . # # # # # # # # . . .
. . . . # . . . . . # . . .
. . . . # . # # # # # . # . . .
. . . . # . # . . . # . # . . .
. . . . # . # . . . # . # . . .
. . . . # . # . . . # . # . . .
. . . . # . # # # # # . # . . .
. . . . # . . . . . # . . .
. . . . # # # # # # # # . . .
. . . . . . . . . . . . . .
. . . . . . . . . . . . . . P
Total ECTS: 56
Enter your move:
YOU WON !
gorkem@gorkem:~/Desktop$
```

YOUTUBE LINK: [https://youtu.be/6LQBzxHM\\_Dk](https://youtu.be/6LQBzxHM_Dk)

Hocam video normal haliyle 4 dakikayı aştığı için biraz hızlandırdım bazı yerler videoda maalesef anlaşılmamış o yüzden raporu olabildiğince detaylı hazırlamaya çalıştım.